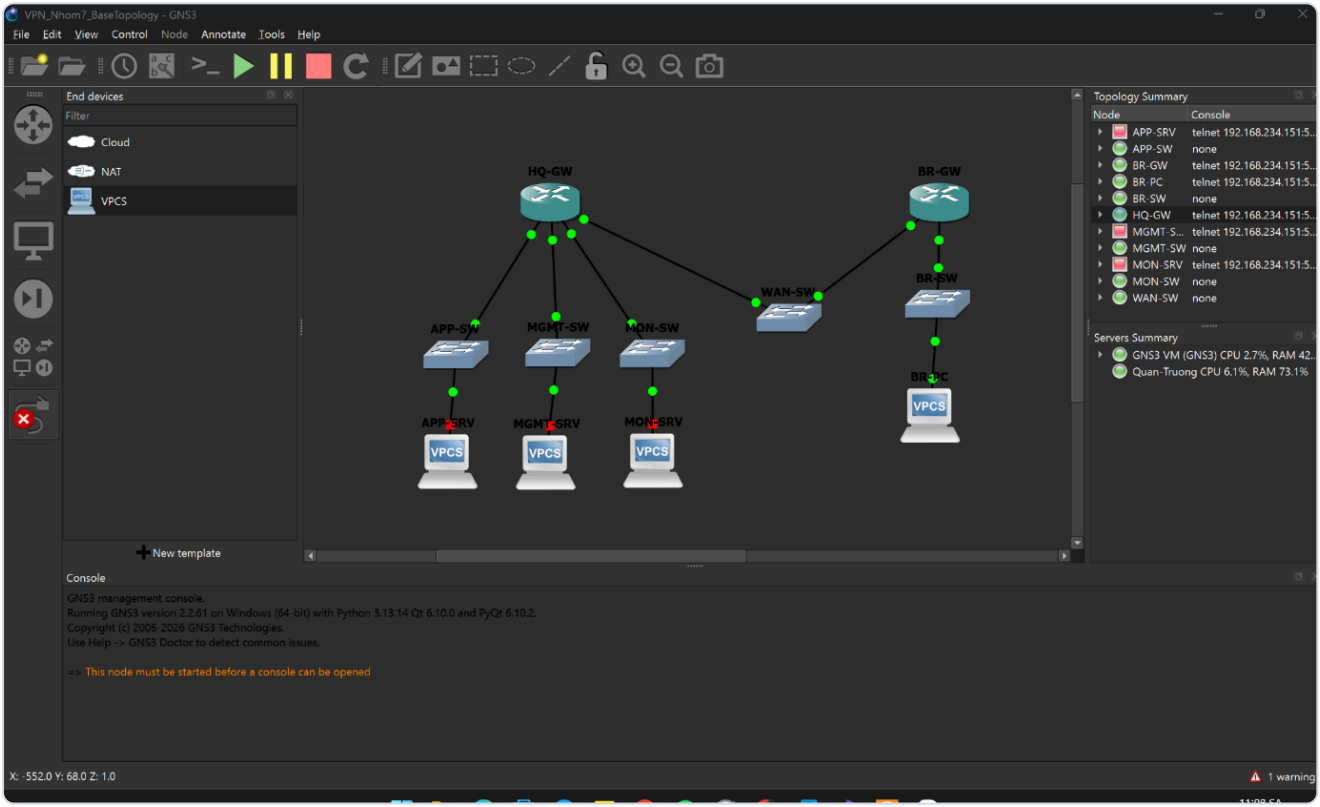
Hybrid VPN Lab

Enterprise network with an HQ, a Branch office and remote users, built with **Docker Compose** (10 containers, 7 networks).

- **Site-to-Site VPN** HQ ↔ Branch: IPsec IKEv2 (strongSwan), AES-256 / SHA-256 / DH Group 14, certificate authentication, DPD, rekey.
- **Remote Access VPN**: WireGuard gateway with two clients, per-peer revocation.
- **Firewall**: nftables, default-deny, APP / MGMT / MON segmentation.
- **Tester**: Kali node on the WAN for scanning and validation.

Topology



IP Plan

Container	WAN (172.16.0.0/24)	Internal	Role
hq-ipsec	172.16.0.10	192.168.60.1	HQ IPsec gateway
br-ipsec	172.16.0.20	192.168.20.1	Branch IPsec gateway
firewall	172.16.0.30	192.168.60.2, 70.2, 10.1, 40.1, 50.1	Zone firewall, gateway of APP/MGMT/MON
wg-gw	172.16.0.100	192.168.70.1, wg0 10.50.0.1	WireGuard gateway
client-a	172.16.0.101	wg0 10.50.0.2	Remote user
client-b	172.16.0.102	wg0 10.50.0.3	Remote user
app	—	192.168.10.10	Application server

mgmt	—	192.168.40.10	Management
mon	—	192.168.50.10	Monitoring
kali-tester	172.16.0.200	—	WAN tester

Networks: `app` 192.168.10.0/24 · `branch` 192.168.20.0/24 · `mgmt` 192.168.40.0/24 · `mon` 192.168.50.0/24 · `hq_transit` 192.168.60.0/24 · `wg_transit` 192.168.70.0/24 · WireGuard tunnel 10.50.0.0/24.

Run

Requires a Linux host with Docker Engine and Docker Compose v2.

```
docker compose up -d --build
```

Quick checks:

```
docker exec hq-ipsec ipsec statusall             # tunnel ESTABLISHED
docker exec app ping -c 4 192.168.20.1           # HQ -> Branch over IPsec
docker exec client-a ping -c 4 192.168.10.10     # remote user -> APP
docker exec client-a ping -c 2 192.168.40.10     # remote user -> MGMT: blocked
docker exec firewall nft list ruleset            # firewall counters
```

Stop: `docker compose down`

Structure

```
compose.yaml   docker/ (images)   configs/ (ipsec, pki, wireguard)   evidence/ (test output,
pcap)
```

Keys and certificates are lab-only.

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